5COSC019W – Object Oriented Programming
Week 1

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5COSC001W - OOP

This module :

- introduces the concepts of objects and classes
- teaches to apply principles of object oriented programming, OOP analysis and design in tackling programming problems
- aims to extend the programming skills you acquired in the first year

Module contents

- Object Oriented Principle and characteristic
- Classes and Objects. The usage of APIs
- Inheritance, polymorphism, abstraction, encapsulation.
- How to design class. Introduction to UML
- Computational modelling
- Graphical User Interfaces
- Concurrency and multithreading
- Defensive programming, Testing, unit testing
- Input/Output and streams
- Fundamental design patterns

Objectives

After this course you should be able to:

- **Tackle** more complicated programming problems
- **Decompose** programming problems using object oriented analysis and design
- Use **most** of the object-oriented programming concepts when writing programs
- **Improve** the knowledge of Java programming language

Learning outcome

- Identify and justify good practices in development of OO software
- Apply acquired knowledge of concepts and programming languages based on OO principles
- Design, implement and test application based on OOP
- Implement GUI interfaces using OOP languages
- Describe, justify apply concurrency principle and how threads can be implemented in an OOP language

Assessments

Consists of :

- **Lab-based Practical** – Coding exercises during seminar slot
- **Exam** – multiple choice questions, fill-in the blanks, analyse code snipped, fill UML diagrams

Lab-based Practical :

- During your seminar in week 11

Exam date:

- January – date TBC

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Plagiarism and Academic Misconduct

"In other words, plagiarism is an act of fraud. It involves both stealing someone else's work and lying about it afterward."

(http://www.plagiarism.org/plagiarism-101/what-is-plagiarism)

- **Do not copy the code from other students**
- **Do not copy the code from external source**
- **You cannot use ChatGPT to generate code**

https://www.westminster.ac.uk/study/current-students/resources/academic-regulations/academic-misconduct

Books and resources

- ***Big Java***, Horstmann, C.   4th edition. Wiley.

Further reading:

- ***Java. A Beginner's Guide***, Herbert Shildt

- ***The object Oriented Thought Process***, 4th Edition, M. Weisfield, Addison Wesley
- ***Design Patterns: elements of reusable object-oriented software***, Gemma, Helm, Johnson, Vlissides. Addison Wesley

My expectations of you

- You attend **all** lectures and tutorials (seminars)

– Lectures are online and the pdf will be available on Blackboard before lecture

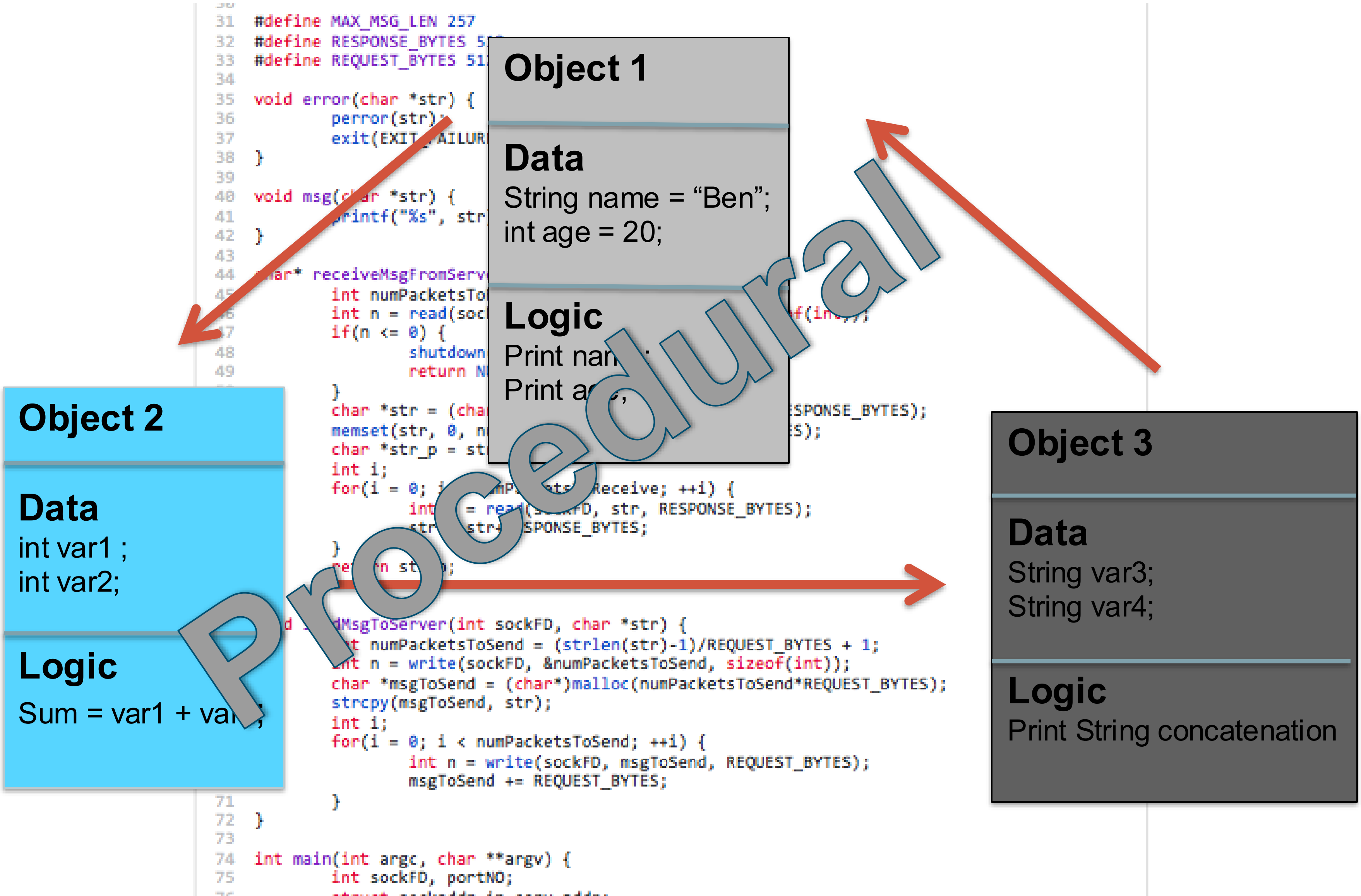
– Tutorials are on-site – look at the timetable for your seminar slot

- Attend all lectures and tutorials **on time**
- You do all tutorial exercises and in-lecture exercises
- Flag any problems you have early

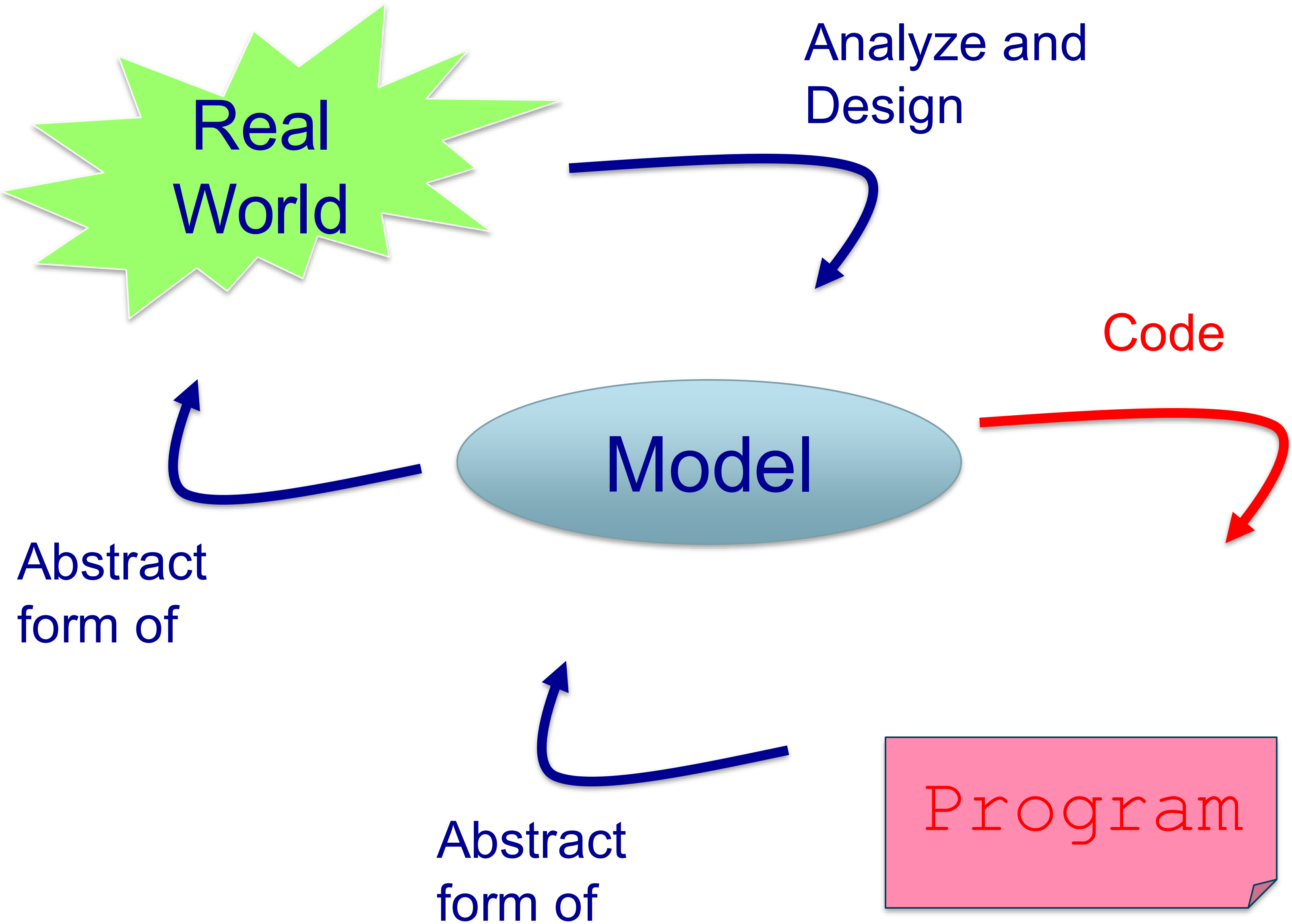
Motivation: Why programming is important?

- Even if you will not become a professional programmer, learning programming is a valuable life skill. It involves:
- ➢ creativity: programming involves problem solving skills...
- ➢ systematic thinking: programming teaches you to reason logically and consequently…
- ➢ collaborative work: coding within a team...

Object Orientation Programming



Model

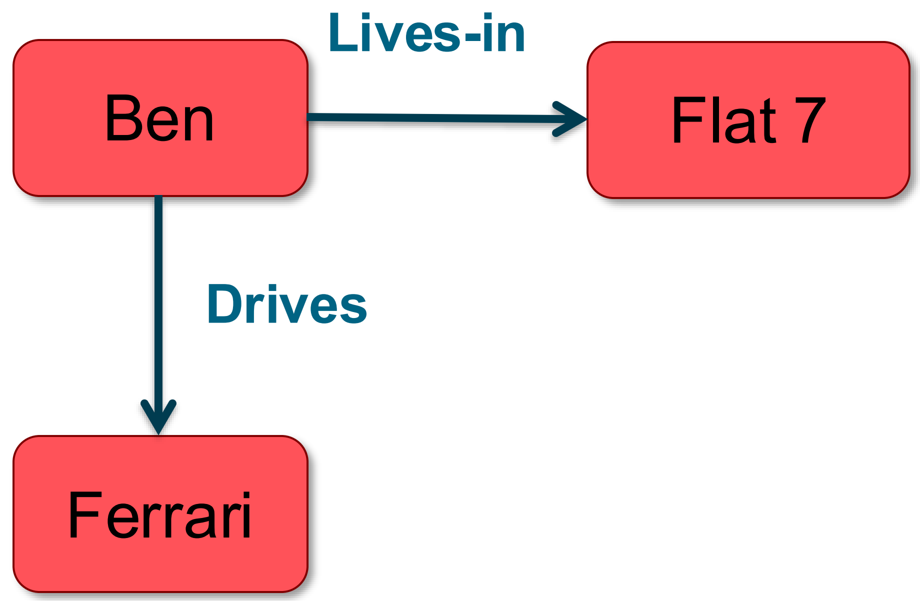

Analyze and
Design
Real
World
Code
Model
Abstract
form of
Abstract
form of
Program

So what is “Object Oriented” about?

- A program typically creates a **model** of a part of the “real” world
- The parts of the model are the **objects** that can be identified in the problem, and these will be included in the software model
- Objects can be categorised into **classes**
- <u>all objects that share similar characteristics or behaviors, that are of the same kind, belong to the same class</u>
- an object that belongs to a class is an **instance** of this class

Example – OO Model

- Objects
  - Ben
  - Flat 7
  - Ferrari
- Interactions
  - Ben lives in the house
  - Ben drives the car



OO - Advantages

- People think in terms of objects
- OO models map to reality
- Therefore, OO models are
  - easy to develop
  - easy to understand

Objects

An object is

- Something tangible (person, pen, mug)
- Something that can be apprehended intellectually (Time, Date)
- An object has **state**, **behavior**, **identity**

Person Object	Person Object	Bank Account Object
Name : “Bob” Age: 22	Name : “Jane” Age: 33	number: 01 23 45 balance: 200 £
speak() walk()	speak() walk()	deposit() withdraw()

Classes

A class is a blueprint or template for creating objects (instances). It defines the structure and behaviour that its instances (objects) will have.

Classes

Type

- Name: What is it?
  - Person, BankAccount, Employee, Time, etc.

Properties, data -> instance variables

- Attributes: What does it describe?
  - Name, Age, Balance, salary, etc.

Operations -> instance methods

- Behavior: What can it do?
  - Speak, deposit, work, etc.

In summary, What are Objects and Classes?

An object is

- Something tangible (Ben, Ferrari)
- Something that can be apprehended intellectually (Time, Date)
- An object is an **Instance of a class.**

    - Ben is an instance of the class Persons: a **specific object** that belongs to the class Persons

- We want our **class** to be a grouping of conceptually-related state and behaviour

Example – Ben is a Tangible Object instance of the class Person:

- State (attributes)
  - Name
  - Age
- Behavior (operations)
  - Walks
  - Eats
- Identity
  - His name

Example – 11:00:00 is a time and is an Object Apprehended Intellectually, it is an instance of the class Time:

- State (attributes)
  - Hours
  - Minutes
  - Seconds
- Behavior (operations)
  - Set Hours
  - Set Minutes
  - Set Seconds
- Identity
  - Would have a unique ID in the model

Our first class

- Suppose you write code for a bank, and you are asked to write a class Account
- What is state, behavior for a bank Account?

- State -> Attributes:
  - Account number
  - balance

- Behaviors -> Operations:
  - Withdraw
  - Deposit
  - Get balance
  - Close account

Representing a Class Graphically (UML)

BankAccount	Name
- balance: double - accountNum: String	State
+ withdraw (amount: double): void + deposit (amount: double): void + getBalance(): double + close(): void	Behaviour

Means Private

Means Public

Example: BankAccount class

BankAccount
- balance: double - accountNum: String
+ withdraw (amount: double): void + deposit (amount: double): void + getBalance(): double + close(): void

Class

BobAcc	**JaneAcc**
15000£ A0123 withdraw() deposit() getBalance() close()	20000£ A0456 withdraw() deposit() getBalance() close()

Object (instance)

The Account Class – Declaration

```
public class Account
{         private double balance;
          private String accountNum;

          public Account(double initialBalance, String accNum)
          { some code }
          public Account()
          { some code }

          public void withdraw(double amount)
          { some code }
          public void deposit(double amount)
          { some code }
          public double getBalance()
          { some code }
          public void close()
          { some code }
}
```

Name of the Class

State: instance variables

Constructors: to initialize objects, to create instances

Behavior: instance Methods, to change or show the state of Account

Class Declaration

- In general:
- By convention a class name always starts with a capital letter.
- A class declaration contains declarations of **instance methods** and **instance variables**
- Typically, a class declaration also contains declarations of **constructors**: their job is to initialise objects

```
public class class_name {
        variable-declarations
        constructor-declarations
        method-declarations
}
```

- The order is not important, but this is a standard way of class declarations

Constructors

- When an object is created, its instance variables are initialised by a **constructor**

```
public class Account
{
        private double balance;
        private String accNumb;

        public Account(double initialBalance, String accountNum)
        { balance = initialBalance;
          accNum = accountNum;
         }
        public Account()
        { balance = 0;}
…
}
```

Constructors

- The constructor **MUST** have the same name as the class name
- A class <u>can have more than one constructor</u>
  - like the example in the previous slides
  - constructor overloading
  - same rules apply as for method overloading

Instance Variables

- An instance variable can hold a single value
  - e.g. `private double balance`;
- But also, an instance variable can hold an object
  - let us not worry about the details of this for now
- Instance variables are typically defined as private
  - this facilitates **information hiding** between classes
  - more on access modifiers (e.g. private) later
  - **information hiding essential in industry**

Instance Methods

- Instance methods represent the behavior and alter the state of an object, they manipulate an object
- In the Account example:
  - ***deposit***: increases the balance of the account by a specified amount
  - ***withdraw***: decreases the balance of the account by a specified amount
  - ***getBalance***: prints the balance of the account
  - ***close***: closes the account

The Account Class

```
public class Account
{
private double balance;
...
public void withdraw(double amount) {
balance -=amount;
}
public void deposit(double amount) {
balance += amount;
}
public double getBalance() {
return balance;
}
public void close() {
 balance = 0;
}
}
```

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Instance Methods

- Instance methods also provide the interface of classes
  - ➢ different objects can communicate with each other via their instance methods
- To achieve this, instance methods are typically defined as `public`
  - ➢ can be accessed by members of other classes

Declaring instance methods

```
public void deposit(double amount)
```

Access modifier

Return type
(void if no value
is returned)

Method name
(you choose the
name)

Parameter list (type
and name). As many
as required

Method Overloading

- Instance methods can be overloaded, i.e. more than one method can have the same name inside the same class
- For this to work, overloaded methods must have different number of parameters, or parameters of different types

```
1) public void deposit(double amount)
   public void deposit(double money)

2) public void deposit(double amount)
   public void deposit(double amount, double interest)
```

Which is valid ???

Access Modifier

- The declaration of an instance variable, a constructor, an instance method (even of a class) begins with an access modifier (public or private)
  - **Public**: the entity can be accessed by other classes
  - **Private**: the entity is only accessible from within the class itself

- In general (there are exceptions) instance variables will be `private`, and instance methods and constructors will be `public`

OOP Principles

Abstraction

Polymorphism

Inheritance

Encapsulation

Abstraction

- Focus on the essential quality of an object
- Discard what is irrelevant
- We create one Class to represent several objects

BankAccount
- balance: double - idNumber: String

15000£ A0123 withdraw() deposit() getBalance() close()

BobAcc

20000£ A0456 withdraw() deposit() getBalance() close()

JaneAcc

Encapsulation

- wrapping the data (variables) and code acting on the data (methods) together as a single unit.
- the variables of a class will be hidden from other classes and can be accessed only through the methods of their current class. Therefore, it is also known as **data hiding**.

Black Boxing

How to do it?

- ➢Declare the variables of a class as private.
- ➢Provide public setter and getter methods to modify and view the variables values.

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The Account Class – get/set methods

```
public class Account
{
private double balance;
private String accountNum
...
public double getBalance()  {
return balance;
}
public String getAccountNum(){
return accountNum;
}
public void setBalance(double newBalance){
balance = newBalance;
}

public void setAccountNum(String newAccountNum){
accountNum = newAccountNum
}
…
}
```

How do we create an object in Java?

- We need to call the constructor of the class in order to create instances of the class
- For this we need the **new** keyword

```
Account account1 = new Account(100);
```

- the above line declares a variable named *account1* of type Account
- it gives to the variable account1 the value *new Account(100)*
- similar to int x =5 where we declare a variable named x of type int and give to the variable x the value 5

Type Account

- *Account* is actually a *new type*.
- This allows *Account* to be used for declarations such as:

```
Account account1= new Account(100);
```

- In fact, Account is a *User Defined Type*.

Call constructors

- We need to call the constructor of the class in order to create instances of the class
- For this we need the new keyword

```
account1 = new Account();
account2 = new Account(100);
```

- ➢ we need to call the constructor with the correct parameters and types
- ➢ remember, we implemented 2 constructors for the Account class in our previous examples

Calling instance method

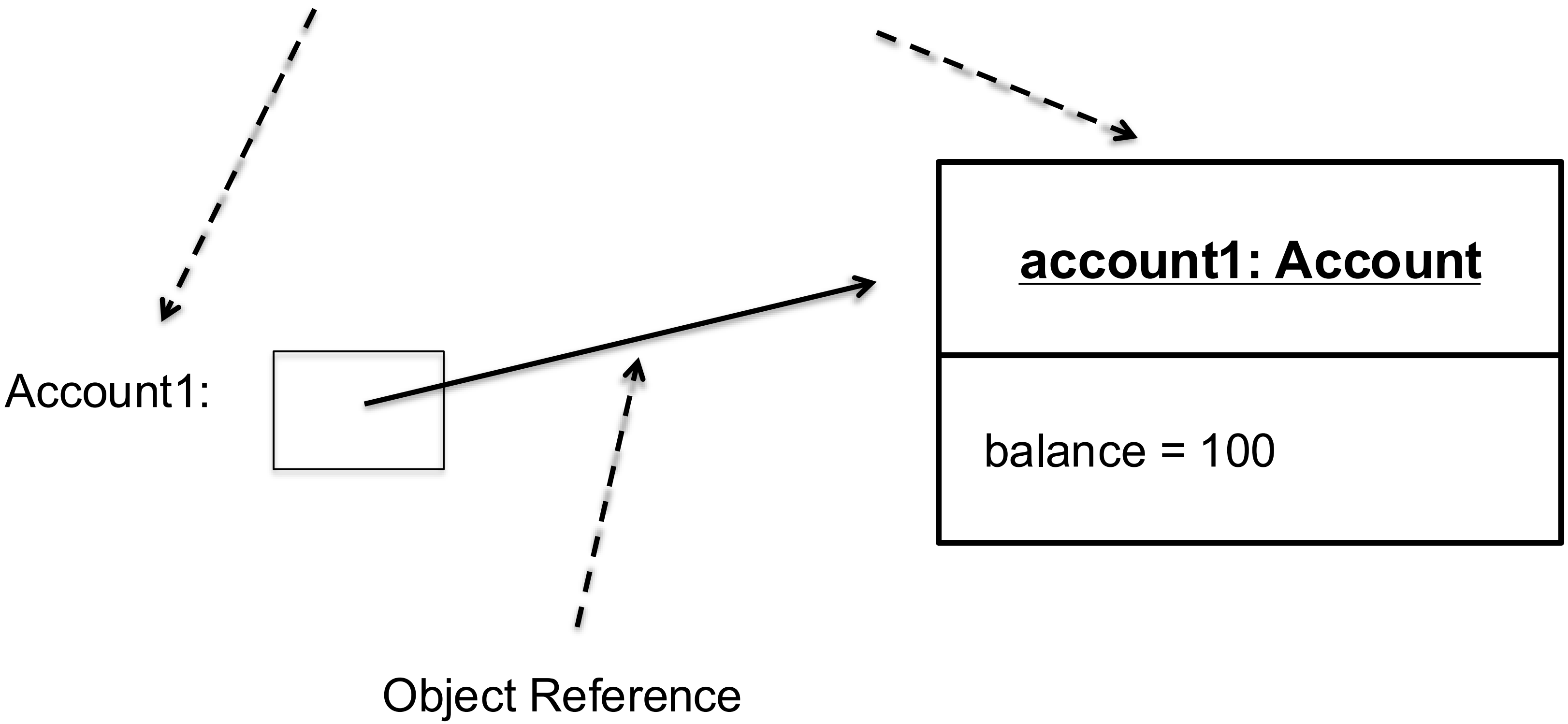
- Once an object has been created, operations can be performed on it by calling the instance methods of the object's class

```
object_name.method_name(arguments)
```

```
account1.deposit(1000);
account2.withdraw(250);
account2.close();
```

Object References

```
Account account1 = new Account(100);
```



Reference

- A variable of a class type holds a *reference* to an object.

– A reference is a pointer (form of memory address).

- Variable doesn’t hold the object itself.
- The variable can go out of scope but the object can *still* exist (providing it is referenced by some other variable).
- One object can be referenced by several references and, hence, variables.

Consequences

- If an object reference is passed as a parameter then:

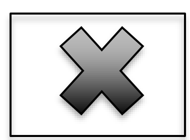
- Changing the object inside the method changes the object outside the method.

- They are the same object!

Null Reference

- `null` keyword.
- No object is referenced, so no methods can be called.
- `Account account1= null;`
- Default value if variable not initialised.

account1:



Summary – What you should know so far…

- Objects and Classes
- Class declaration
- Object as instance of a class
- Constructors
- Object Assignment
- Overloading methods
- Values and References